

The answer is yes. ASCO has one global optimiser (DE) and two local optimisers (Hooke&Jeeves and Nelder-Mead). The configuration file contains parameters to tune the DE whereas the source code allows the tuning of local optimisers. Thus the two local optimisers can be selected only at the time of compilation by editing the file 'asco.c'

DE generates new parameter vectors by adding the weighted difference vector between two population members to a third member. In situations where the resulting vector yields a lower objective function value than a predetermined population member, the newly generated vector replaces the vector it is compared to.

Distributed computation methods

Currently, ASCO supports the following three methods for parallel computation:

Send. Each process receives one vector at a time and returns the single cost back to the Master process before receiving any new vector.

Scatter. All the vectors are scattered among all the existing processes in the same generation, and all evaluations are performed before returning all the cost values back to the Master process, which is repeated in the next generation.

Scatter with load balancing. In situations where the power of the computer is evenly available to each process, the optimisation time decreases. This is either due to different machines or due to machines with different loads balancing the number of simulations according to the computational resources. However, in perfectly balanced situations, all the processes start and finish the optimisation at the same time.

Competition

ASCO is one of its kind but has plenty of competition when it comes to circuit optimisation. Although all those mentioned below are not optimisers, they do have in-built algorithms to support optimisation as a feature.

TINA. Though TINA is a design suite for circuit simulation and PCB design, it has in-built circuit optimisation capability apart from analysis, designing and real-

How to install and use ASCO

Typing the following commands at command prompt will help create the executables for installation:

```
tar -zxvf ASCO-<version>.tar.gz
cd ASCO-<version>
make
```

This will create two executables 'asco' and 'asco-test' that need to be copied to a common directory so that these can be used later.

However, for portability building and for installing ASCO across different environments, GNU Autotools are recommended. If you are using Autotools, you only need to type the following command at the command line:

```
tar -zxvf Autotools.tar.gz
```

All the necessary files like configure.ac and Makefile.am are packed in Autotools.tar.gz and are distributed along with ASCO. In order to configure and compile further, the following commands are required:

```
aclocal
automake -f -c -a
sh autogen.sh
./configure
make
```

To use ASCO, copy the executable ASCO to examples/<YOUR SIMULATOR>/inv. If you are using Eldo, execute the following commands:

```
cp asco examples/Eldo/inv
cd examples/Eldo/inv
./asco -eldo inv
```

If you are using any other simulator, replace Eldo and eldo with appropriate text.

Following is the link to the binary version of ASCO for win32:

http://asco.sourceforge.net/downloads_win32.html

ASCO can also be compiled in win32 using 'cygwin' and 'MinGW32.' However, in case of any issues 'Makefile.win32' can be used.

time testing of digital, analogue, HDL, MCU and mixed electronics circuits. It has Berkeley Spice and XSpice based SPICE simulator engines. It refines the result of a design procedure and tunes the already working circuits.

DELIGHT.SPICE. It is a computer-aided design (CAD) system and a SPICE circuit analysis program with interactive optimisation. It contains the recent and powerful optimisation algorithms along with a methodology for the designers for their intuition and man-machine interaction.

Utmost IV. It is used to generate accurate and compact models, macro models and Verilog-A models for RF

and analogue applications. It simulates as well as optimises any combination of data, including the data values that have been extracted. It supports SPICE simulators HSPICE, SmartSpice, Spectre and Eldo. Utmost IV combines direct extraction and parameter optimisation techniques.

EvoSpice. It is a numerical optimiser for LTSpice IV. It maximises or minimises the circuit behaviour aspects (power dissipation, output amplitude, etc) by automatically selecting the component values. It uses rapid evolution instead of differential evolution and is hundred times faster as compared to the other systems.

A member of electro-tech-online says, "ASCO requires you to compile C code and therefore is a bit difficult to use whereas EvoSpice is available only as a limited free trial. Anyway, both can be used to automatically adjust component values in a simulation, to minimise or maximise some aspect of a circuit's performance."

Scope and audience

ASCO requires an experienced circuit designer to select the circuit topology and to find reasonable operating conditions. The fact is that it's an experienced designer who can define the test benches and measurements to achieve the desired design objectives, define realistic design goals and evaluate the circuit of the proposed sizes to select the most suitable one.

One of the key requirements of using this tool is that, the experienced designer should be well versed with the operation of a SPICE simulator. The designer is expected to at least understand the consequences of each of the assumptions made during optimisation to carry out the optimisation process efficiently.

Surprisingly, despite being an optimiser for the popular simulation tools from SPICE, ASCO has not been much in use! There are some questions regarding the usage of this tool but no reviews to find out how useful this tool has been. ●

The author is a technical journalist at EFY. The article is based on the documents available at <http://asco.sourceforge.net/>